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The Effects of ADHD on Human Pathology

Attention deficit hyperactivity disorder, or ADHD, is a common neurological disorder. ADHD is a complex, lifelong disease, although not completely understood, that causes problems with paying attention, controlling impulses, and hyperactivity, mostly from chemical imbalances in the brain. In the United States specifically, it is one of the most frequently diagnosed childhood psychiatric disorders, and is so common that even I, Owen Santiago, have it (CDC, "Attention-Deficit or Hyperactivity Disorder (ADHD)"; National Institute of Mental Health, "Attention-Deficit or Hyperactivity Disorder (ADHD)").

So, I sat down and started writing about what ADHD is, to discover how it affects people in their day-to-day lives, and how it goes beyond the usual symptoms that people know about. I thought it would be exciting to see why I am the way I am. Why can I sit down and get an hour's worth of work done in 20 minutes, or sit for an hour and get only ten minutes of work done? Why do I feel like I'm going to jump out of my skin and run around the room when I'm feeling super excited? ADHD has affected my life in many different ways, so many that it's hard to count them all. And so here I am, explaining to you what people with ADHD deal with on a day-to-day basis and how it affects people with ADHD beyond just its basic symptoms. But what is ADHD?

ADHD has three different types: ADHD inattentive, ADHD hyperactive or impulsive, and ADHD combined. ADHD inattentive primarily causes a lack of attentiveness but doesn't usually cause hyperactivity. ADHD's symptoms, despite common misconceptions, are NOT due to defiance or lack of comprehension, or the newer generation's lack of focus. Rather, it's because the chemical composition of the ADHD brain isn't tuned quite right. ADHD inattentiveness, in my own experience, makes you feel a little tired, and people with ADHD inattentive often go on tangents.

Alternatively, ADHD hyperactive-impulsive causes a person to be primarily hyperactive and impulsive, but doesn't usually cause someone to be unattentive. They are often restless and have constant movement, and are usually susceptible to constant fidgeting, tapping, and talking. These are usually the very energetic kids on the playground, and the ones with the pencil that's always wiggling or with the pen that's always clicking. These symptoms are, again, due to a chemical imbalance in the brain.

However, the most common form of ADHD, ADHD combined, causes a person to be hyperactive and inattentive. People with ADHD combined have the whole package: They often act without thinking, interrupt others, have difficulties with self-control, desire IMMEDIATE rewards, and the inability to delay gratification, along with the symptoms of the hyperactive-impulsive and inattentive forms of ADHD (CDC, "Attention-Deficit or Hyperactivity Disorder (ADHD)"). This is the kind I have, and it has significantly affected me, causing me to struggle to pay attention in school, miss important events in my life, and strain relationships. It's all due to ADHD combined. But what can cause someone to "get ADHD?"

ADHD is not caused by one thing or another. Unlike a viral, bacterial, or fungal infection, people with ADHD get their disease at birth, and have it for the rest of their lives. The only way you can pass it on is if you're predisposed to have it. These predispositions are caused by a variety of factors, including genetics, improper development of the central nervous system, and other prenatal and postnatal factors.

ADHD is predominantly a genetic disease that it runs in families (The Editors of Encyclopaedia Britannica). For example, if "You have blood relatives, such as a parent or sibling, with ADHD or another mental health disorder," the risk of ADHD may increase (Mayo Clinic). Some studies even estimate the heritability of ADHD to be roughly 70–90%, indicating a connection to the underlying metabolic reactions, directly or indirectly involving gene products (Mayo Clinic; Cickovski et al.). Improper development of the central nervous system is also a major cause of ADHD. (CHADD, "The Science of ADHD - CHADD"). Commonly caused by environmental exposure during childhood to substances such as lead and toxins, improper development of the central nervous system can cause children to develop ADHD due to the resulting changes in the brain (Mayo Clinic).

Other potential risk factors can occur prenatally and postnatally. Prenatally, a premature birth (pre-37th week of pregnancy), a low birth weight, maternal smoking, alcohol use, and/or drug use during pregnancy can cause ADHD. Postnatally, brain damage, either in the womb or after a severe head injury, can cause ADHD (as one would expect after brain trauma) (CDC, "Attention-Deficit or Hyperactivity Disorder (ADHD)"; National Institute of Mental Health, "Attention-Deficit or Hyperactivity Disorder"; National Institute of Mental Health, "Attention-Deficit or Hyperactivity Disorder"; CHADD, "The Science of ADHD - CHADD"; Pan and Bölte; Vasconcelos et al.; Jansone et al.; Stray et al.).

Although ADHD is a neurological disorder, it affects - or is related to - other diseases. For example, the dopaminergic dysfunction caused or related to ADHD has some significant effects on other parts of the body. Notably, ADHD is possibly related to the same neurobiological and genetic mechanisms that cause obesity, which are related to the release of dopamine in the brain (Instanes et al.). Similarly, ADHD's central nervous system dopaminergic dysfunction is also linked to sleep problems, restless leg syndrome, migraines, and, in conjunction with inflammation, asthma. Generalized joint hypermobility (GJH) has also been observed in association with ADHD, potentially due to shared genetic causes affecting connective tissue and the central nervous system (Glans et al.). Celiac disease has been found in patients with ADHD symptoms as well (Instanes et al.). But this is only the tip of the iceberg of ADHD's effect on the entire body.

ADHD is fundamentally a neurodevelopmental disorder, affecting how the brain grows and develops, so the nervous system is a great place to start (CDC, "Attention-Deficit or Hyperactivity Disorder (ADHD)"). Generally, ADHD's most significant effect is the neurotransmitter imbalances, which are the chemical messengers that transmit signals between nerve cells. There are two main victims: dopamine and norepinephrine (CDC, "Attention-Deficit or Hyperactivity Disorder (ADHD)").

Dopamine is the neurotransmitter that causes feelings of reward, motivation, and pleasure ("Inside the ADHD Brain: Structure, Function, and Chemistry"; The Editors of Encyclopaedia Britannica). For example, if you studied for a test and then got a good grade on it, dopamine would be produced, creating a relationship between studying and getting a good grade. People with ADHD have lower dopamine levels, which makes it harder for people with ADHD to be motivated, especially for tasks that offer delayed or smaller rewards, like doing chores or doing a

job and waiting to get paid. As a result, people with ADHD prefer immediate gratification over long-term rewards, which therefore leads to the impulsive behavior associated with ADHD (ADDA Editorial Team, "Inside the ADHD Brain: Structure, Function, and Chemistry").

Norepinephrine, meanwhile, is the neurotransmitter involved in attention, alertness, and executive functioning (ADDA Editorial Team, "Inside the ADHD Brain: Structure, Function, and Chemistry"). Particularly important in the prefrontal cortex, where executive (high-level) functions occur in the human brain, norepinephrine acts as an inhibitor, preventing people from ADHD to focus on the things that matter and to suppress impulses ("Inside the ADHD Brain: Structure, Function, and Chemistry"; Stray et al.; Abraham et al.). ADHD's disruption of norepinephrine transmission as a result makes it hard for people with ADHD to retain information, suppress distractions, and control impulses (ADDA Editorial Team, "Inside the ADHD Brain: Structure, Function, and Chemistry").

MRI scans have shown that there are physical structural differences throughout the brains of people with and without ADHD (The Editors of Encyclopaedia Britannica). The frontal cortex, responsible for executive functions, behavior regulation, attention, and planning, is affected by ADHD, maturing at a slower pace or showing disrupted activity and connectivity (ADDA Editorial Team, "Inside the ADHD Brain: Structure, Function, and Chemistry"). The basal ganglia, which control motor control, behavior regulation, and planning, are also disrupted in people with ADHD, affecting motor learning, focus, and multitasking ability (ADDA Editorial Team, "Inside the ADHD Brain: Structure, Function, and Chemistry"). Even the hippocampus and amygdala, which process memory and emotional processing, respectively, are found to be different in size and shape, leading to notable effects on memory and emotional processing (ADDA Editorial Team, "Inside the ADHD Brain: Structure, Function, and Chemistry"). The

cerebellum, which is related to emotional, impulse control, and attentional processes, is also affected by ADHD, further reducing focus (Stray et al.). It also may play a role in the lack of attention and cognitive function in folks with ADHD (Stray et al.).

The many interconnected neurons in the brain create a complex network of neurotransmitters. Two networks are the Default Mode Network and the Frontostriatal Network, which are both significantly affected by ADHD (Cickovski et al., "Inside the ADHD Brain: Structure, Function, and Chemistry").

The Default Mode Network (DMN), which is active during mind wandering and daydreaming, is more active in people with ADHD, causing people to have a harder time focusing on tasks (ADDA Editorial Team, "Inside the ADHD Brain: Structure, Function, and Chemistry"). Meanwhile, the Frontostriatal Network, which connects the frontal cortex with the basal ganglia and is involved in executive functions, attention, and motor control, is affected by ADHD (Stray et al.). ADHD can disrupt the connectivity and activity in this network, causing more of the core ADHD symptoms (ADDA Editorial Team, "Inside the ADHD Brain: Structure, Function, and Chemistry"). The effect that ADHD has on these neural networks causes people with ADHD to struggle to focus and complete tasks, in conjunction with the pre-existing neurotransmitter differences.

People with ADHD also have electroencephalography (EEG) abnormalities (Jansone et al.). Scientists used an EEG, which measures brain wave activity, to see the differences in the brain's electrical patterns in people with ADHD (Jansone et al.). The main differences are found in the increased ratio between the slow-wave activity (theta waves, which are involved in memory consolidation and restoration) to fast-wave activity (beta waves, which are involved in

cognitive processing and sensory input), notably in the frontal cortex of the brain (Jansone et al.). As a result, people with ADHD do more thinking and input and less memory consolidation and remembering, leading to more inattentiveness.

Medications can affect the neurological side of ADHD significantly ("Attention-Deficit or Hyperactivity Disorder"; The Editors of Encyclopaedia Britannica). Stimulants like methylphenidate (Ritalin) and amphetamine (Adderall) increase the levels of dopamine and norepinephrine in the brain ("Attention-Deficit or Hyperactivity Disorder", The Editors of Encyclopaedia Britannica). Alternatively, non-stimulants like atomoxetine (Strattera) primarily work by increasing norepinephrine levels.

ADHD affects the muscle system via the neurological differences in the ADHD brain, causing problems with motor regulation (Stray et al.). Two common symptoms associated with ADHD are motor inhibition problems and muscle tone issues.

Motor inhibition problems cause people with ADHD to struggle suppressing unwanted movements, such as the rapid repetition of simple movements, like diadochokinesis (pronation-supination movement of one hand with the elbow flexed at a 90-degree angle) and thumb movements, leading to jerky, uncoordinated, and effortful movements, particularly during tasks requiring repetitive actions (Stray et al.). Relatedly, people with ADHD also experience heightened muscle tone or a muscle's resting tension, causing restricted movement in the shoulders, hips, vertebral column, and thorax. Hip movement specifically was restricted by 80%, according to a study by Liv Larsen Stray, Øistein Kristensen, Martha Lomeland, Mette Skorstad, Torstein Stray, and Finn Egil Tønnessen (Stray et al.). This increased muscle tone may cause shortness of breath due to restricted respiration, caused by a lack of thorax movement and limited

leg and foot flexibility and balance, resulting in clumsiness in people with ADHD, as well as widespread chronic pain & fatigue due to constant elevated muscle tension (Stray et al.).

ADHD is possibly also linked to people with generalized joint hypermobility (GJH). This link “may represent a marker for an underlying systemic disorder involving both connective tissue and the central nervous system” (Glans et al.).

Methylphenidate (MPH), a stimulant medication that I use, causes many of these symptoms to decrease over time at the expense of a small but statistically significant increase in heart rate (Stray et al.; Vitiello). This is a noticeable effect of this medication, as I have also noticed that, while my heart rate is higher on average, I am much less tense while I’m on my medication compared to when I’m off it.

Skeletally, people with ADHD tend to have chronic musculoskeletal and skeletal complaints, such as fibromyalgia, which causes widespread musculoskeletal pain that causes fatigue, sleep, memory, and mood issues (Instanes et al.; Stray et al.). These symptoms go hand and hand with the heightened muscle tone of people with ADHD, as the higher muscle tension causes there to be more stress, and therefore more wear on the skeleton and muscles, leading to fibromyalgia (Instanes et al.; Stray et al.). These symptoms are often misdiagnosed as somatization disorder, or the condition where a person experiences physical symptoms that are out of proportion to their actual medical condition (Glans et al.).

Adults with ADHD also tend to have generalized joint hypermobility, or GJH, which causes a person to have an unusually large range of motion in their joints (Glans et al.). Despite the researchers Martin Glans, Nils Thelin, Mats B. Humble, Marie Elwin, and Susanne Bejerot adjusting their tests for comorbid anxiety disorder, the most common associated disease among psychiatric conditions, GJH persisted in their study (Glans et al.). This enhanced range of motion also causes pain similar to fibromyalgia, as well as the continued wear on their bones. Thus, GJH is a co-occurring condition with ADHD. ADHD causes significant effects on the musculoskeletal system, with enhanced muscle tone and generalized joint hypermobility causing fibromyalgia and general musculoskeletal pain.

Considering the integumentary system, people with ADHD, especially children, are more likely to have atopic dermatitis, or eczema (Instanes et al.; Zhou et al.). A study in Turkey found that, in 60 patients with eczema and 50 without, patients with eczema displayed significantly more ADHD symptoms than those without, with the most common occurrences being in women (Instanes et al.; Glans et al.). Worst of all, individuals with ADHD are twice as likely to have acne compared to other dermatological disorders (Instanes et al.; Zhou et al.). These might be due to shared genetic or environmental factors (Pan and Bölte, Zhou et al.).

Again, generalized joint hypermobility (GJH) affects another system, the skin. GJH causes “un- usually soft or velvety skin, mild skin hyperelasticity, and atrophic scarring” (Glans et al.). But with adult ADHD, GJH with additional musculoskeletal symptoms and/or skin abnormalities is more common (Glans et al.). As a result, the co-occurrence of generalized joint hypermobility causes people with ADHD to have velvety skin and musculoskeletal pain, like fibromyalgia.

Generally, ADHD itself doesn't affect the circulatory system. Rather, medications used to treat ADHD have more side effects. ADHD medications, such as stimulants and atomoxetine, can increase heart rate and blood pressure, though not to unhealthy levels. Concerns that this can cause sudden cardiac death exist, however, are unfounded, as no conclusive evidence has linked stimulant use and SCD (Vitiello).

Besides medications, however, a study by Lin Li, Zheng Chang, Jiangwei Sun, Miguel Garcia-Argibay, Ebba Du Rietz, Maja Dobrosavljevic, Isabell Brikell, Tomas Jernberg, Marco Solmi, Samuele Cortese, and Henrik Larsson determined that ADHD can cause individuals to be more susceptible to various cardiovascular diseases after an average of 11.8 years of follow-up from their first cardiovascular incident. The study specifically adjusted for factors such as education level, lifestyle choices (smoking, diet), and other health conditions (diabetes, obesity, sleep problems), but the results were conclusive: ADHD can cause an individual to be subject to "increased risk was present across all types of cardiovascular diseases, but the strength of the associations was greatest for cardiac arrest, hemorrhagic stroke and peripheral vascular disease/arteriosclerosis" (Li et al.).

ADHD has almost zero effect on the lymphatic system, which circulates lymph, the fluid that helps maintain fluid balance and supports the immune system by defending against infections and diseases. However, there are a few effects on the general immune system. Research has found a link between ADHD and immune system dysregulation, as people with ADHD are more susceptible to neurological problems due to inflammation, like asthma, as one study by Pei-Yin Pan & Sven Bölte found a significant connection between the two (Pan and Bölte.). Inflammatory diseases like asthma are common in people with ADHD, as are allergic

diseases like atopic eczema and allergic rhinitis (Instanes et al.; Zhou et al.). Hypotheses for why this happens include:

- Elevated inflammatory cytokines (inflammatory chemicals) released during allergic responses could cross the blood-brain barrier and affect brain development, resulting in ADHD, according to a Chinese study by Rong-Yi Zhou, Jiao-Jiao Wang, Ji-Chao Sun, Yue You, Jing-Nang Ying, Xin-Min Han (Zhou et al.).
- Genetic factors crisscross with ADHD and these diseases (Pan and Bölte; Zhou et al.).
- Prenatal stress (a significant risk factor for ADHD) may cause the development of allergies (Zhou et al.; Eng et al.).

Research also indicates a potential connection between ADHD and higher levels of the pro-inflammatory cytokine TNF- β , along with reduced levels of anti-inflammatory cytokines like IL-4, IL-2, and IFN- γ . Also, maternal inflammatory and immune system diseases during pregnancy have been linked to an increased risk of ADHD in offspring (Zhou et al.). Overall, this shows how people with ADHD are more susceptible to allergic reactions, possibly triggered by prenatal stress during pregnancy, and that the chemicals created by these reactions can affect brain development and cause children to have ADHD.

In the respiratory system, asthma is common in people with ADHD. In the Norwegian Prescription Database, there is a higher-than-expected occurrence of ADHD in individuals being treated for asthma (Instanes et al.). Both have shared risk factors, and both start in childhood and co-occur with other psychiatric disorders like mood & anxiety disorders. Ironically, there is a

higher chance of smoking in individuals with ADHD (Instanes et al.; "Adult Attention-Deficit or Hyperactivity Disorder (ADHD) - Symptoms and Causes"; Jansone et al.; "Coexisting Conditions - CHADD"). Both are involved in the dopaminergic system. There are dopamine receptors found in airways, and dopamine may have a bronchodilating effect, particularly during asthma attacks (Instanes et al.; "Attention-Deficit or Hyperactivity Disorder (ADHD)."; "Inside the ADHD Brain: Structure, Function, and Chemistry"; Pan and Bölte; Budziszewska et al.; The Editors of Encyclopaedia Britannica).

ADHD may be related to the immune system, along with asthma. Some theories propose that inflammatory cytokines, released during allergic reactions, might interfere with brain development in regions crucial for attention and executive function (Instanes et al.; Zhou et al.). ADHD has also been observed to have high comorbidity with Th1- and Th2-mediated disorders such as ear infections, eczema, and asthma (Zhou et al.). One study suggests that the increased muscle tone of ADHD could restrict chest movement, especially with the increased muscle tone of chest muscle (Stray et al.).

ADHD may have a pathogenesis linked to the endocrine system - the body's hormone system - specifically in the HPA (hypothalamic-pituitary-adrenal) axis, which is what manages stress (Kozłowska et al.; Budziszewska et al.). Studies have found weaker HPA axis activation indicators due to psychological stress, namely from lower stress hormone (i.e. cortisol) levels, abnormal cortisol circadian rhythms, and lack of cortisol suppression by dexamethasone (Budziszewska et al.). This dysregulation of the HPA axis leads to excessive glucocorticoid (like cortisol) exposure during fetal and early postnatal development, leading to permanent changes in how the brain's glucocorticoid receptor levels (Budziszewska et al.). These glucocorticoids -

which are a type of steroid hormones - cause enhanced dopamine system activity, perhaps leading to the reduced sensitivity to reward in people with ADHD (Budziszewska et al.).

Another pathogenesis of ADHD may be related to thyroid hormones, as “attention and motor activity disorders often occur in children with generalized resistance to thyroid hormones” (Budziszewska et al.). However, studies have found no significant changes in the levels of triiodothyronine (T3) and thyroxine (T4), which are crucial for regulating the body's metabolism, influencing heart rate, body temperature, growth, and development, nor thyroid-stimulating hormone (TSH), which regulates the thyroid gland's production of T3 and T4 (Budziszewska et al.; Instanes et al.).

Androgens, a group of hormones that cause male development and reproduction that include testosterone, may have a big role in the origination of ADHD, at least in men. Men with ADHD may be more sensitive to testosterone due to the decrease in blood flow to the frontal cortex. This itself is due to the reduction of estrogen receptor-alpha and vascular endothelial growth factor (VEGF) levels. These factors cause men with ADHD to have disturbances in memory processes. Androgens, however, are poorly understood in their relation to ADHD, so the findings made by Budziszewska et al. may not be significant.

A study by Vasconcelos, Mônica Maria de Almeida et al. found that people with ADHD are 2.3 times more likely to have lower urinary tract symptoms (LUTS), more commonly in the combined subtype. This could possibly be caused by the relationship between “noradrenergic and dopaminergic pathways in the central nervous system, [the result of which is] ...decreased adrenergic activity affecting the lower urinary tract," which leads to bed-wetting (enuresis) (Vasconcelos et al.). Enuresis is common in individuals with ADHD, but its association with

ADHD is not an agreed-upon related symptom. For instance, Johanne Telnes Instanes et al. conducted research showing how one study from 1999 to 2001 found no significant difference in the prevalence of enuresis between adults with ADHD and a control group when using a US healthcare claims database. Meanwhile, another study from 2006 shows that there is a significant increase in the prevalence of enuresis among adults with ADHD. Additionally, a study by Vasconcelos et al. found that inattention and hyperactivity-impulsivity in early childhood were associated with higher odds of enuresis at age 10 and in adolescence (Vasconcelos et al.).

In the reproductive system, medications used by people with ADHD cause the most harm. One study by Pia Brandt Danborg, Anders Lykkemark Simonsen, and Peter C. Gøtzsche found that, in mammals, methylphenidate and clonidine impaired reproduction and may cause irreversible harm to animals if treatment isn't started early in life. The study did not find effects on the number of stillborns, the number of offspring, or visible malformations. However, one study found that the pelvis, pubis, and sternum were incompletely ossified in newborns compared to controls.

Treatment with these medications can decrease the occurrence of precocious puberty with sexual hyperactivity. One study found that a boy who was treated with methylphenidate for 17 years, starting in childhood, experienced delayed puberty. At age 20, he had underdeveloped genitalia, a lack of libido, and erectile dysfunction (Danborg et al.).

ADHD has an extensive diagnosis and treatment regimen. ADHD does not have any recognizable physical symptoms and cannot be diagnosed with a simple test, and instead requires a comprehensive evaluation ("Diagnosing ADHD - CHADD"; The Editors of Encyclopaedia Britannica). The process involves clinical evaluation, medical examination (including hearing

and vision tests to see if other medical conditions may be causing ADHD-like symptoms), and questionnaires completed by the patient, family, and teachers ("Diagnosing ADHD - CHADD"; "Attention-Deficit or Hyperactivity Disorder (ADHD).")

The diagnostic criteria for ADHD are outlined in the Diagnostic and Statistical Manual of Mental Disorders (DSM), of which the current version is the DSM-5 (CHADD, "Diagnosing ADHD - CHADD"). A diagnosis is only considered appropriate and necessary if the disturbance is not judged to be the direct pathophysiological consequence of another medical condition ("Diagnosing ADHD - CHADD"; "Coexisting Conditions - CHADD"). To meet the diagnostic criteria of ADHD, the symptoms must be chronic or long-lasting, impair functioning, and cause a person to fall behind typical development for their age. The symptoms must also be present in two or more settings (e.g., at home, school, work, with friends) (National Institute of Mental Health, "Attention-Deficit or Hyperactivity Disorder"). The specific criteria differ slightly for children and adults. "Children up to 16 years must show at least six symptoms of inattention, hyperactivity and impulsivity, or both." (National Institute of Mental Health, "Attention-Deficit or Hyperactivity Disorder"). In contrast, adults and youth over 16 years old must show at least five symptoms of inattention, hyperactivity, and impulsivity, or both. (National Institute of Mental Health, "Attention-Deficit or Hyperactivity Disorder").

Multiple kinds of professionals can diagnose children with ADHD. These professionals include clinical psychologists, clinical social workers, nurse practitioners, neurologists, psychiatrists, pediatricians, primary health care providers, child psychologists, child psychiatrists, and developmental pediatricians (for children) ("Diagnosing ADHD - CHADD"; CDC; "Attention-Deficit or Hyperactivity Disorder"). Meanwhile for adults, the diagnostic process may involve ADHD symptom checklists, standardized behavior rating scales, detailed

history of past and current functioning, information is obtained from family members or significant others who know the person well, and an evaluation of the history of childhood problems in behavior and academic domains (CHADD, "Diagnosing ADHD - CHADD").

ADHD, as one of the most common neurological disorders, has many treatments available. The primary treatment for people with ADHD is medication, while behavioral therapy, psychotherapy, and counseling can help people with ADHD change their behavior for the better. School accommodations and programs can also be used to help people with ADHD deal with and manage their symptoms. People with ADHD can also be treated by creating an organizational structure, as this can help people with ADHD create routines and organize their lives in a way that is less affected by their impulsivity.

Medications to treat ADHD fall into two categories: stimulant and nonstimulant (National Institute of Mental Health, "Attention-Deficit or Hyperactivity Disorder"). Stimulant medications like methylphenidate (i.e. Ritalin) and amphetamine (i.e. Adderall) increase levels of dopamine and norepinephrine, which are both neurotransmitters that are involved in thinking and attention ("Attention-Deficit or Hyperactivity Disorder"; The Editors of Encyclopaedia Britannica; "The Science of ADHD - CHADD"). Despite being stimulants - and controlled substances - the stimulating effect of these medications helps to fill in the deficit for these substances in the ADHD brain, helping people with ADHD be calmer and focused (The Editors of Encyclopaedia Britannica). They are thus very effective, working for 80% of people with ADHD - a statistic I am a part of, as I take Ritalin (McCabe). However, there are some cardiovascular side effects, as well as decreased appetite, involuntary tics, headache, irritability, and insomnia (Instanes et al.; Vitiello; Budziszewska et al.; The Editors of Encyclopaedia Britannica). Mood swings, hyperactivity, or fatigue can also occur as the medication's effects

wear off. These are considered safe under medical supervision, as these medications are considered Schedule II controlled substances due to their high potential for abuse from their highly stimulating nature (National Institute of Mental Health, “Attention-Deficit or Hyperactivity Disorder”).

Nonstimulant medications are also very common, with medications including atomoxetine (Strattera) and bupropion (The Editors of Encyclopaedia Britannica). These medications take longer to show effects, but also can improve focus, attention, and impulsivity (National Institute of Mental Health, “Attention-Deficit or Hyperactivity Disorder”). They often work by inhibiting the reuptake of norepinephrine, ensuring that there is enough norepinephrine in the brain for normal function (The Editors of Encyclopaedia Britannica). These medications are often the secondary choice, usually taken when stimulants are ineffective, cause unwanted side effects, or are being used in combination with stimulants to enhance their effectiveness (The Editors of Encyclopaedia Britannica). Antidepressants might also be used if a person has co-occurring conditions like anxiety or depression (The Editors of Encyclopaedia Britannica).

Various forms of therapy and counseling are crucial to combating ADHD symptoms and improving overall function. To start, behavioral therapy aims to help people change their behavior - the name is self-explanatory. Behavioral therapists provide practical assistance (ex. assisting in organizing tasks or completing schoolwork, offering guidance on working through hard things), which is immensely helpful for people with ADHD, who tend to struggle organizing the chaos in their minds (“Attention-Deficit or Hyperactivity Disorder”; “Attention-Deficit or Hyperactivity Disorder (ADHD).”). It also helps people with ADHD and their parents, especially during preschool (4-5 years), when parents who are considering diagnosis can see if medication is really necessary (National Institute of Mental Health,

“Attention-Deficit or Hyperactivity Disorder”). School-based behavioral interventions may be required, and constitute the standard behavioral management plans and classroom-taught organizational and study skills (National Institute of Mental Health, “Attention-Deficit or Hyperactivity Disorder”). Behavioral therapy generally causes people with ADHD to monitor and give awareness to their behaviors, and has parents and other authority figures praise them for good behavior, making people - especially children with ADHD - be better at controlling anger, or thinking before acting (National Institute of Mental Health, “Attention-Deficit or Hyperactivity Disorder”). Behavioral therapy is often paired with stimulants and produces the best results.

Meanwhile, psychotherapy consists of many types of therapies. Cognitive behavioral therapy (CBT) can help people to be aware of and accept their thoughts and feelings, improving focus and concentration (National Institute of Mental Health, “Attention-Deficit or Hyperactivity Disorder”). This helps people with ADHD after treatment, as it can help people with ADHD take time to think before acting on things (National Institute of Mental Health, “Attention-Deficit or Hyperactivity Disorder”). Meanwhile, social skills training is just that - teaching people interpersonal and social skills, such as waiting for their turn and asking for help (in children), as well as communication skills like active listening and social problem-solving (in adults). People with ADHD also tend to struggle with interpreting facial expressions and tone of voice.

Families with ADHD also tend to struggle maritally and with relationships, due to the poor communication and disruptive behaviors of ADHD. Thus, marital and familial therapy can help families deal with disruptive behaviors and create positive change for all the members in the family (National Institute of Mental Health, “Attention-Deficit or Hyperactivity Disorder”). Parenting skills training and support groups are also useful for families with ADHD, helping

parents teach skills to encourage and reward positive behaviors in their ADHD children, as well as gain solutions to similar problems (National Institute of Mental Health, “Attention-Deficit or Hyperactivity Disorder”). Stress management training, relatedly, is also very helpful, helping parents deal with the frustration of impulsivity, poor memory, and hyperactive behavior that are constantly present in children with ADHD (which I can also relate to; I was a little ball of energy and incessant talking with zero memory as a kid) (National Institute of Mental Health, “Attention-Deficit or Hyperactivity Disorder”). Life coaches and professional therapists can also be huge tools for people with ADHD, helping people with ADHD organize their lives via routines, lists, calendars, reminders, exercise, and alarms - all of which I personally use to great success (National Institute of Mental Health, “Attention-Deficit or Hyperactivity Disorder”).

Organization is also key to ADHD treatment, as people with ADHD struggle to keep things straight in their heads - why should they, if they're often hyperactive and impulsive, and/or unable to concentrate? Routines, especially in my own experience, have changed my life from one of constant forgetfulness to one where order at least *exists* - impulsivity still gets in the way of *completing* the tasks, which is why counseling is important. Tasks need to be broken down to be more efficient and manageable, which helps combat a huge component of ADHD - procrastination.

ADHD is a complex disease, with a complex diagnostic process, and possibly even more complex treatment. Nevertheless, ADHD is a lifelong condition, and it requires the help of many professionals, from psychologists to pharmacists to therapists, to be able to get medication and support, and to be truly able to manage a condition that can affect people's lives quite significantly.

In the case study “Creative Thinking in Adolescents with Attention Deficit Hyperactivity Disorder”, conducted by Anna Abraham, Sabine Windmann, Rainer Siefen, Irene Daum, and Onur Güntürkün, researchers investigated how the widened attentional focus typically associated with ADHD can enhance creative ability, as commonly postulated. The case study looks at adolescents with ADHD, those with conduct disorder (a mental health condition that causes children and teenagers to display a pattern of aggressive and antisocial behaviors), and a healthy control sample. All of them had similar IQ scores and were about the same age (~ 13.48), so no one group had any advantages or disadvantages besides their disorder or condition.

Generally, people with ADHD did show some “selective cognitive advantages and disadvantages by demonstrating an enhanced ability in overcoming the constraining influence of examples, but a reduced capacity to generate a functional invention during an imagery task” (Abraham et al.). These tests were done with a number of different “challenges.”

The first challenge was called Conceptual Expansion - essentially, the researchers asked the subjects to go wild. The prompt they were given was “animal on another planet.” The farther from normal it was, the more creative they were. For example, did they draw an animal with no legs, or three heads, or glowing eyeballs?

The next was the Creative Imagery Task. Participants were given a creative construction challenge, given 3 random shapes, like a cone, sphere, and cross, and combine them into something new - like a vehicle or house. They were graded on originality and practicality. For example, a cross with balls for wheels and a cab shaped like a cone is creative, but impractical.

Their third test was the Recently Activated Knowledge Task. The participants were told to create a new toy. Before starting, they were shown three toy examples, all containing a ball, high physical activity, and electronics. Then, researchers saw how much those three elements were incorporated into the participants' toy designs. The more like them they were, the more influenced they were by the examples to be as creative and novel as possible.

The final task was the Alternate Uses Task. Here, the goal was to see what the participants could come up with for everyday objects like a newspaper, shoe, or brick. The results were measured on fluency (number of uses participants thought of) and uniqueness (how rare or unusual the thoughts were.)

The results were as the researchers expected. The results showed that there were "significant differences" in the Recently Activated Knowledge task (Abraham et al.). The ADHD group was more creative and less likely to include the common elements present in the toy examples, and the ADHD group scored higher on the originality in the creative imagery test. However, there were some lower trends when it came to practicality, showing that their inventions may have been more creative, but less functional or usable. These results show that, generally, this shows how impulsivity plays a major role in the ADHD mind, at least in the inattentive and combined types.

Overall, ADHD is a complex disorder that affects the entire body, with many origins, symptoms, and effects. While it can present challenges in some areas, it's not all negative. In fact, ADHD can empower individuals to think outside the box and create things no one else has imagined, even if those ideas aren't always the most practical. The benefits it brings can often counteract the detriments. It's even speculated that some of history's greatest geniuses, like

Albert Einstein, Isaac Newton, Walt Disney, and Leonardo DaVinci, may have had ADHD alongside high-functioning autism - well, besides Mr. Disney. Ultimately, ADHD touches many lives and has contributed unique perspectives, ideas, and creations to the world.

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